Frequentist concerns in the land of the Bayesians -- evidence for the statistical anisotropy of the Universe

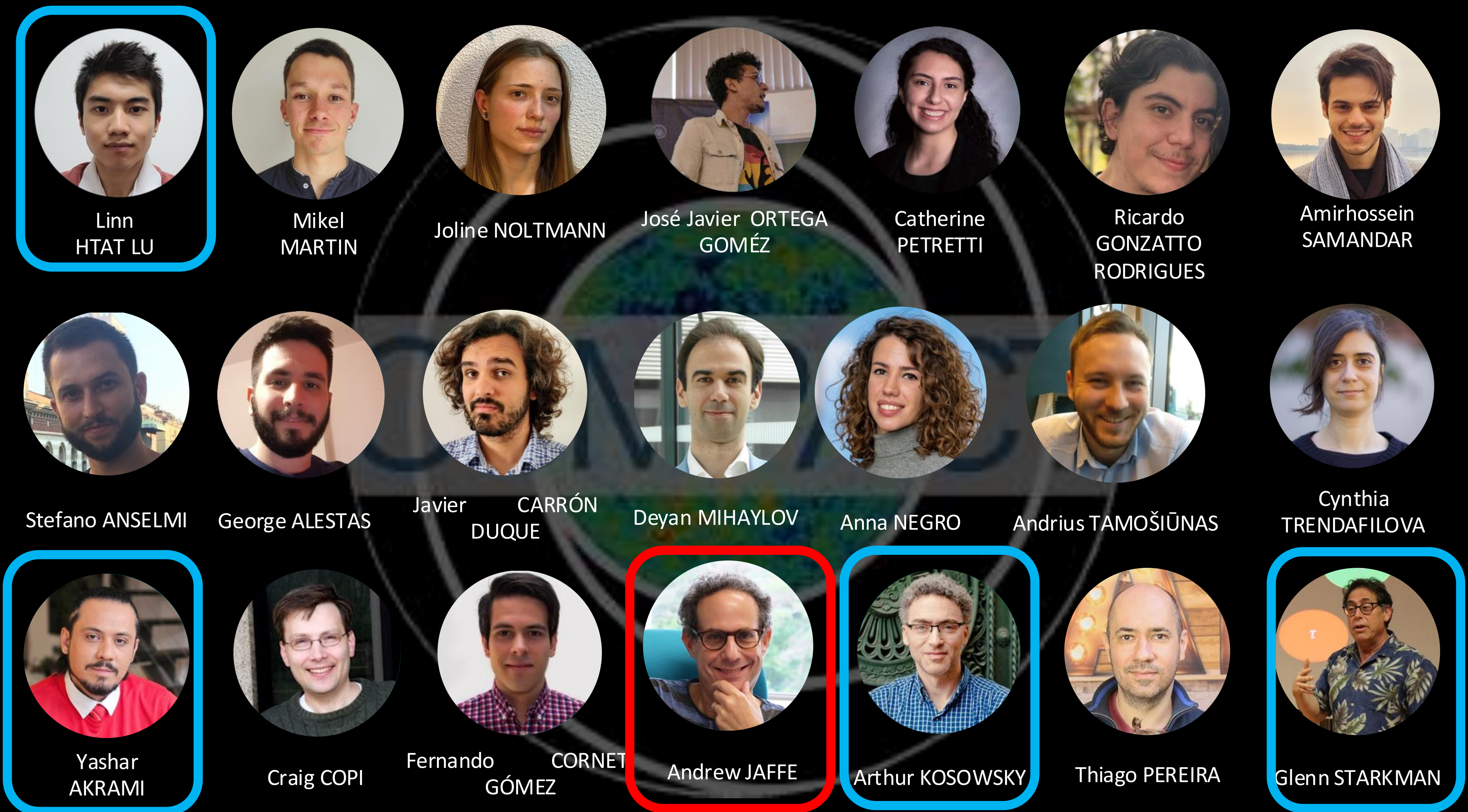
Glenn Starkman

postdoc colleague (CITA) 1993-4!

Case Western Reserve University

The Random Universe (Happy Birthday Andrew!)

ICL June 2026


Linn
HTAT LU
Mikel
MARTIN
Joline NOLTMANN
José Javier ORTEGA
GOMÉZ
Catherine
PETRETTI
Ricardo
GONZATTO
RODRIGUES
Amirhossein
SAMANDAR
Stefano ANSELMI
George ALESTAS
Javier CARRÓN
DUQUE
Deyan MIHAYLOV
Anna NEGRO
Andrius TAMOŠIŪNAS
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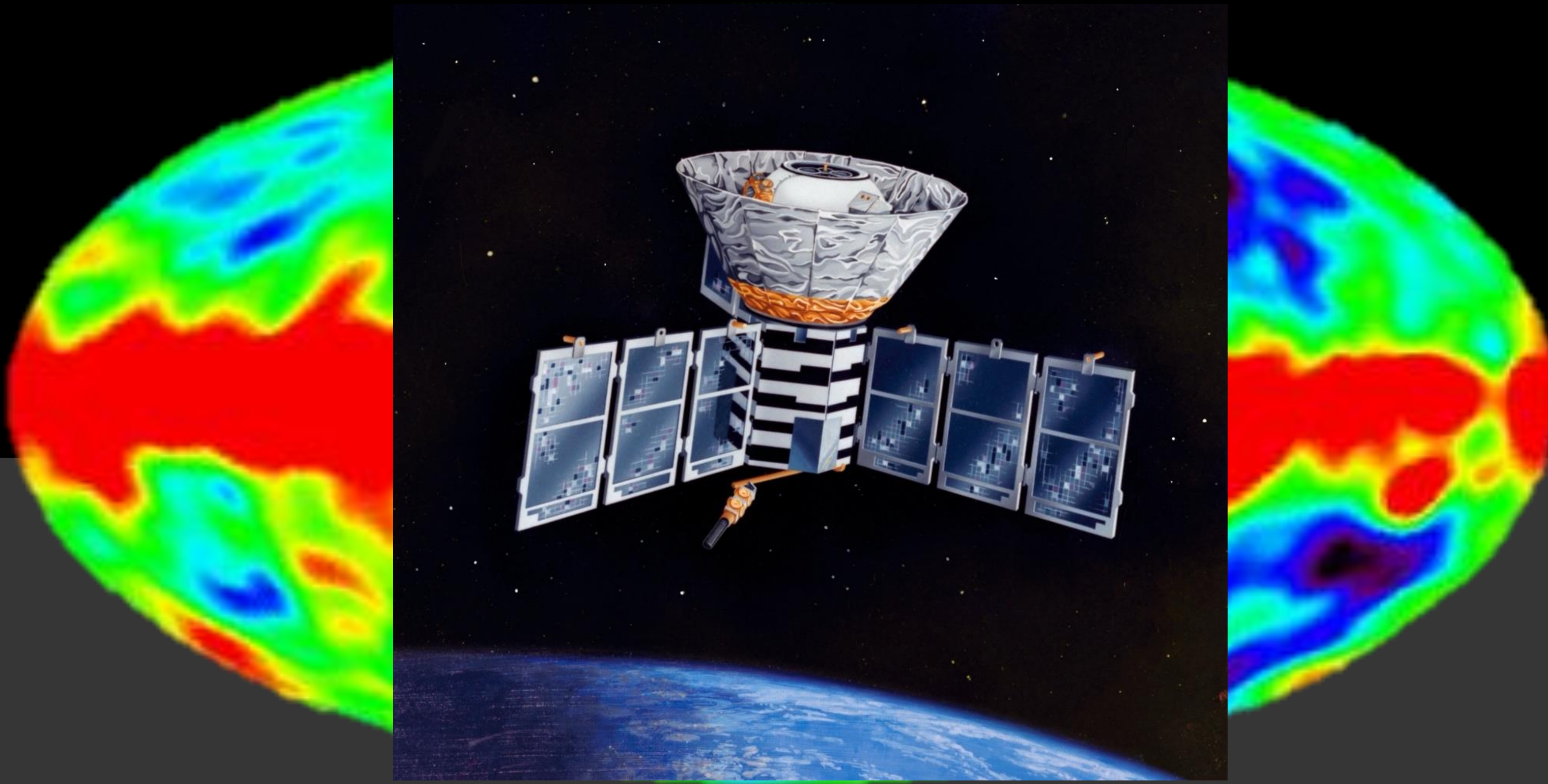
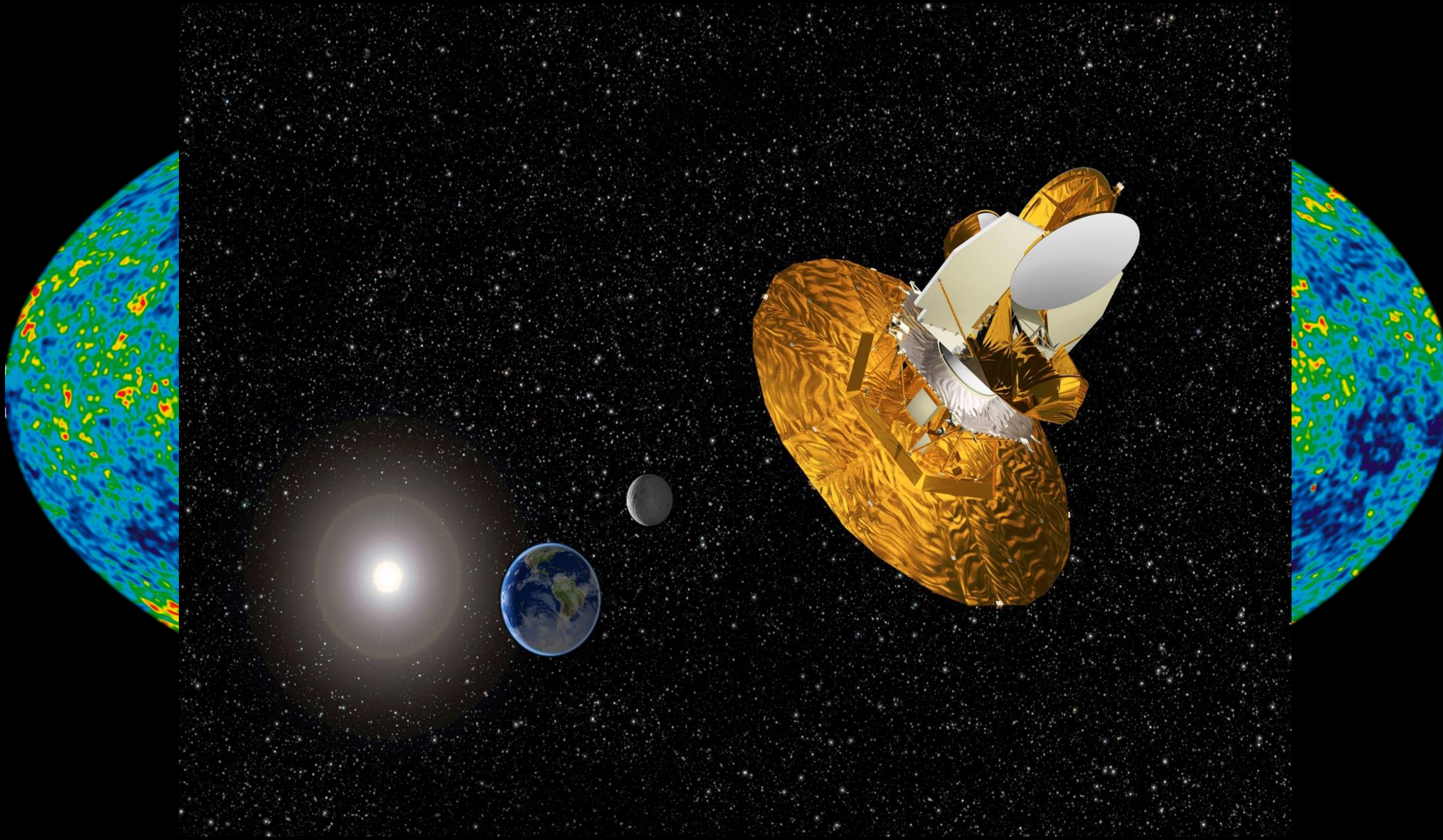
COBE - DMR



NASA/COBE-DMR science team

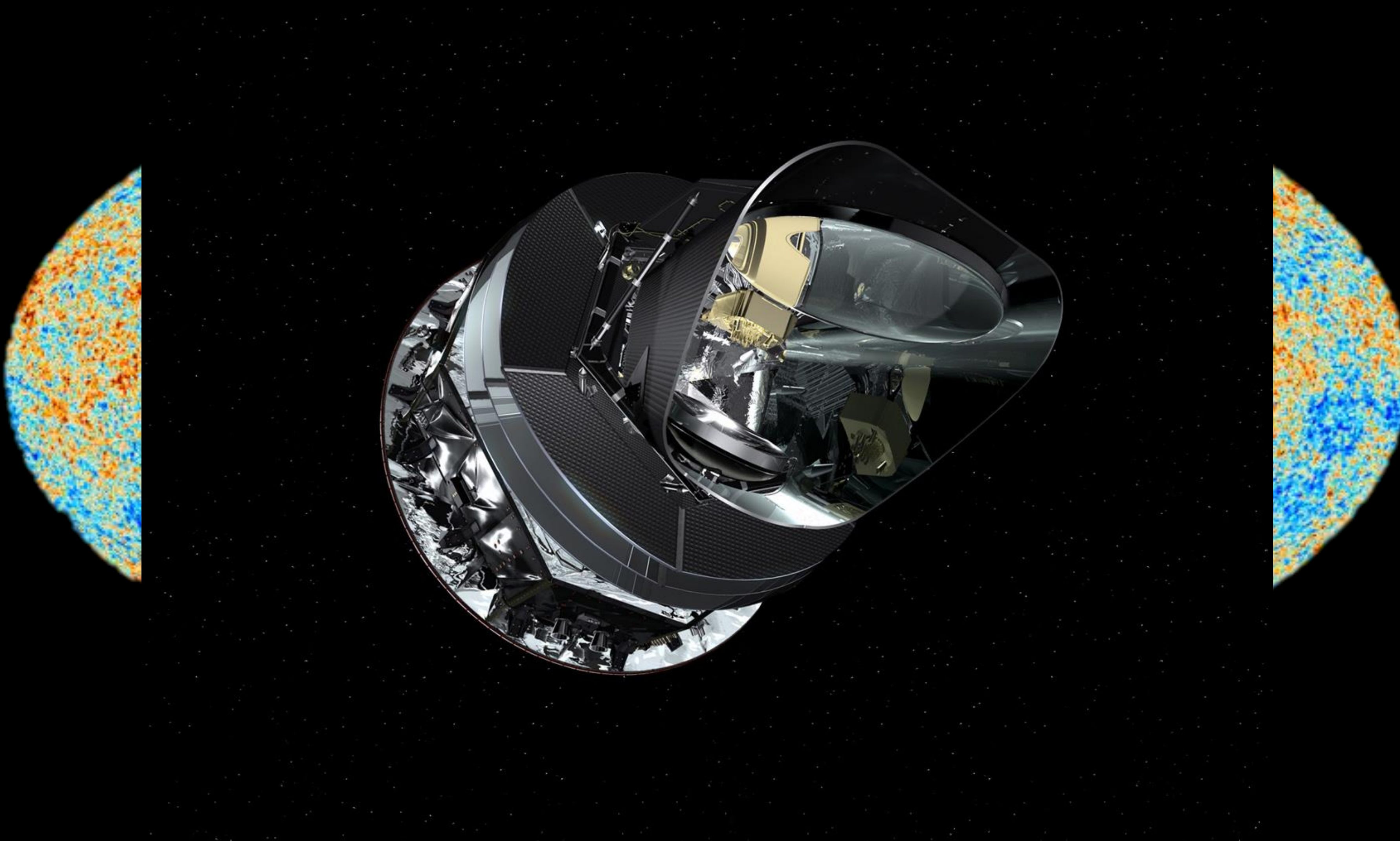
T=18μK

WMAP



NASA/WMAP Science team

Planck



ESA/Planck Science team

ACT

SPT

SO

Angular Power Spectrum

$$\Delta T = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\theta,\varphi)$$

Angular Power Spectrum

$$\Delta T = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\theta,\varphi)$$

Standard model for the fluctuations (inflation):

- Sky is statistically isotropic
- $a_{\ell m}$ are independent Gaussian random variables

$$< a_{\ell m} a^*_{\ell' m'} > = C_\ell \delta_{\ell \ell'} \delta_{mm'}$$

ALL(~) interesting information is contained in:

$$\hat{C}_\ell = (2\ell + 1)^{-1} \sum_m |a_{\ell m}|^2$$

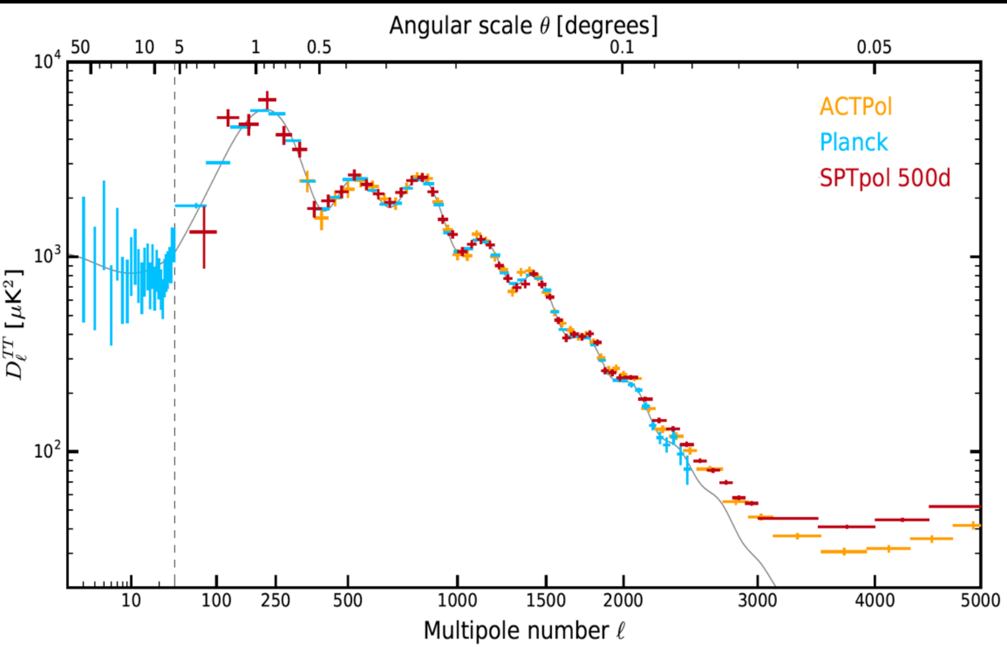
Cosmological Principle

Universe is homogeneous and isotropic
– at least statistically!

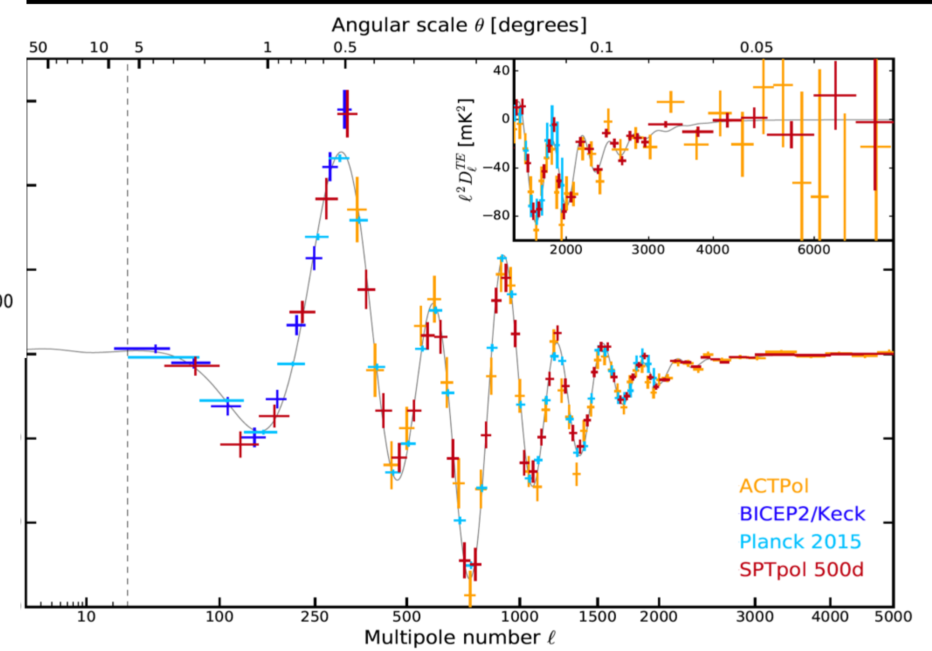
What has the Cosmological Principle done for us?

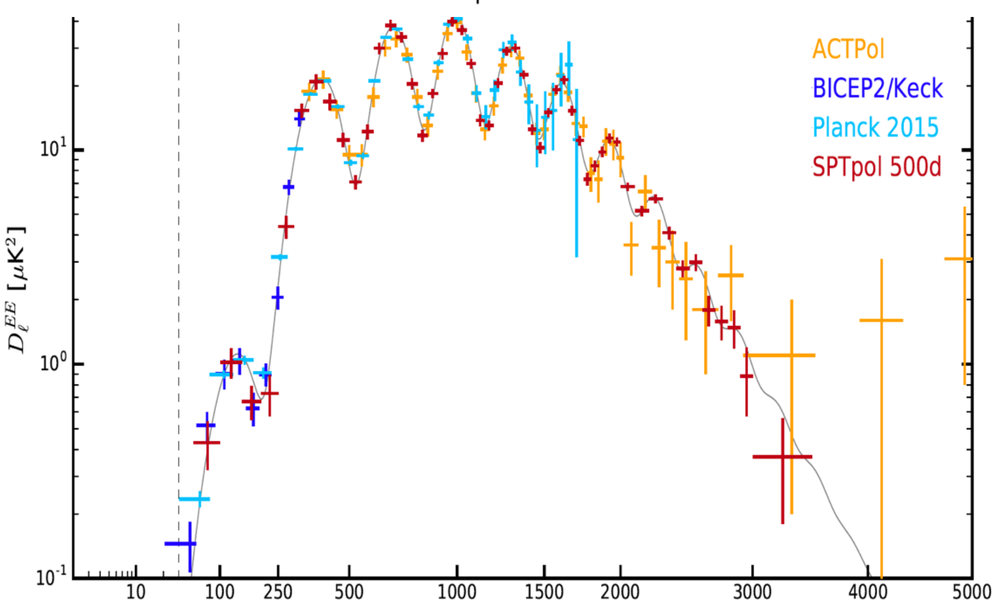
- vast simplification of Einstein equations to two Friedmann equations for background
- organize perturbations in representations of the rotational symmetry
- analyze key non-equilibrium processes in (homogeneous) cooling universe
- replace ensemble averages of statistical quantities with sample averages

Angular Power Spectrum



6 (really 7) parameter fit
to >>7 points





SPTPol arXiv:1707.09353

Astonishing

experimental accomplishment

Remarkable

agreement with theory

BUT !

Standard Model for fluctuations (inflation):

- Sky is statistically isotropic
- $a_{\ell m}$ are independent (very nearly) Gaussian random variables

$$\langle a_{\ell m} a^*_{\ell' m'} \rangle = C_\ell \delta_{\ell\ell'} \delta_{mm'}$$

ALL interesting information is contained in: C_ℓ

Shouldn't we check?!

Shouldn't we check?!

Andrew (~2005):

What do you mean by "check"?

Do you have an alternative theory?

Principle of Bayesian purity:

No theory [that incorporates your statistics in the likelihood, then] no [basis for interpreting your] "facts"

Outline

Troubles in (iso)tropical paradise:

- large-angle problem: $C(\theta > 60^o) \simeq 0$
- low-ℓ alignments
- hemispheres
- parity

Bottom line:

>5σ case against GRSI LCDM realization

in ways that demand statistical anisotropy.

topology musings – left to Yashar, Linn

The first hint: "The low-ℓ Anomaly"

THE ASTROPHYSICAL JOURNAL, 403:L51–L54, 1993 February 1

AN ESTIMATE OF THE *COBE* QUADRUPOLE

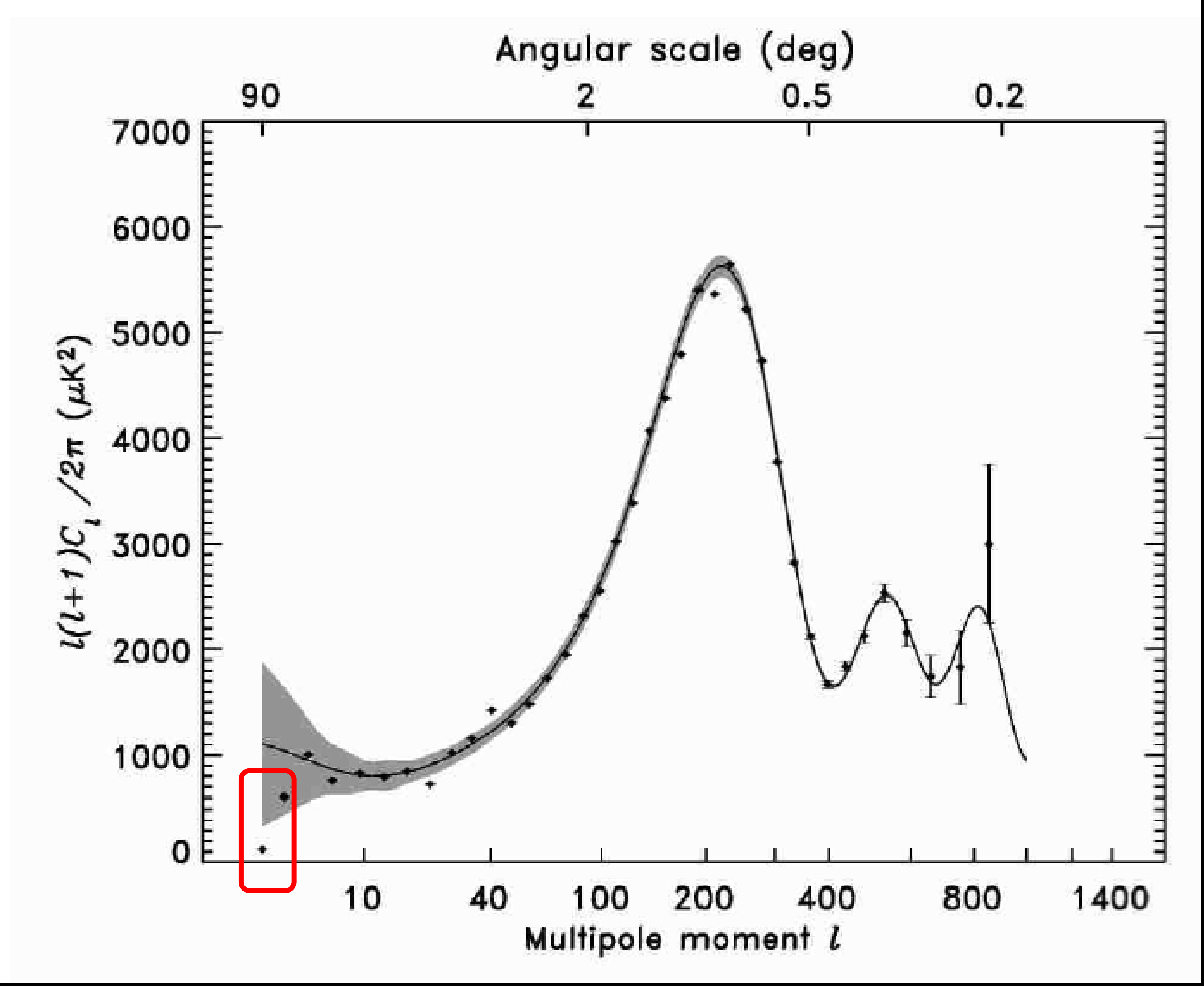
ANDREW GOULD
Institute for Advanced Study, Princeton, NJ 08540
Received 1992 July 7; accepted 1992 November 10

ABSTRACT

Bennett et al. and Smoot et al. reported in 1992 the first positive measurement of the fluctuations in the cosmic microwave background, including a 3 σ detection of the quadrupole, $Q_{\rm rms} = 13 \pm 4$ μK (as inferred from the individual quadrupole components in the "combination" and "reduced galaxy" $|b| > 10°$ maps). I show that part of the observed fluctuations are actually due to noise, so that the best estimate of $Q_{\rm rms}$ in these maps is 10.5 μK. This estimate of $Q_{\rm rms}$ is substantially below the value $Q_{\rm rms-PS} \sim 16$ μk predicted for the quadrupole by extrapolating to $l = 2$, the observed power-spectrum index ($n = 1.1$), from higher multipole moments. This reduced estimate of the quadrupole helps explains why the best fit to the spectral index steepens to $n = 1.5$ when the quadrupole is included in the fit.

Subject heading: cosmic microwave background

The first hint: "The low-ℓ Anomaly"



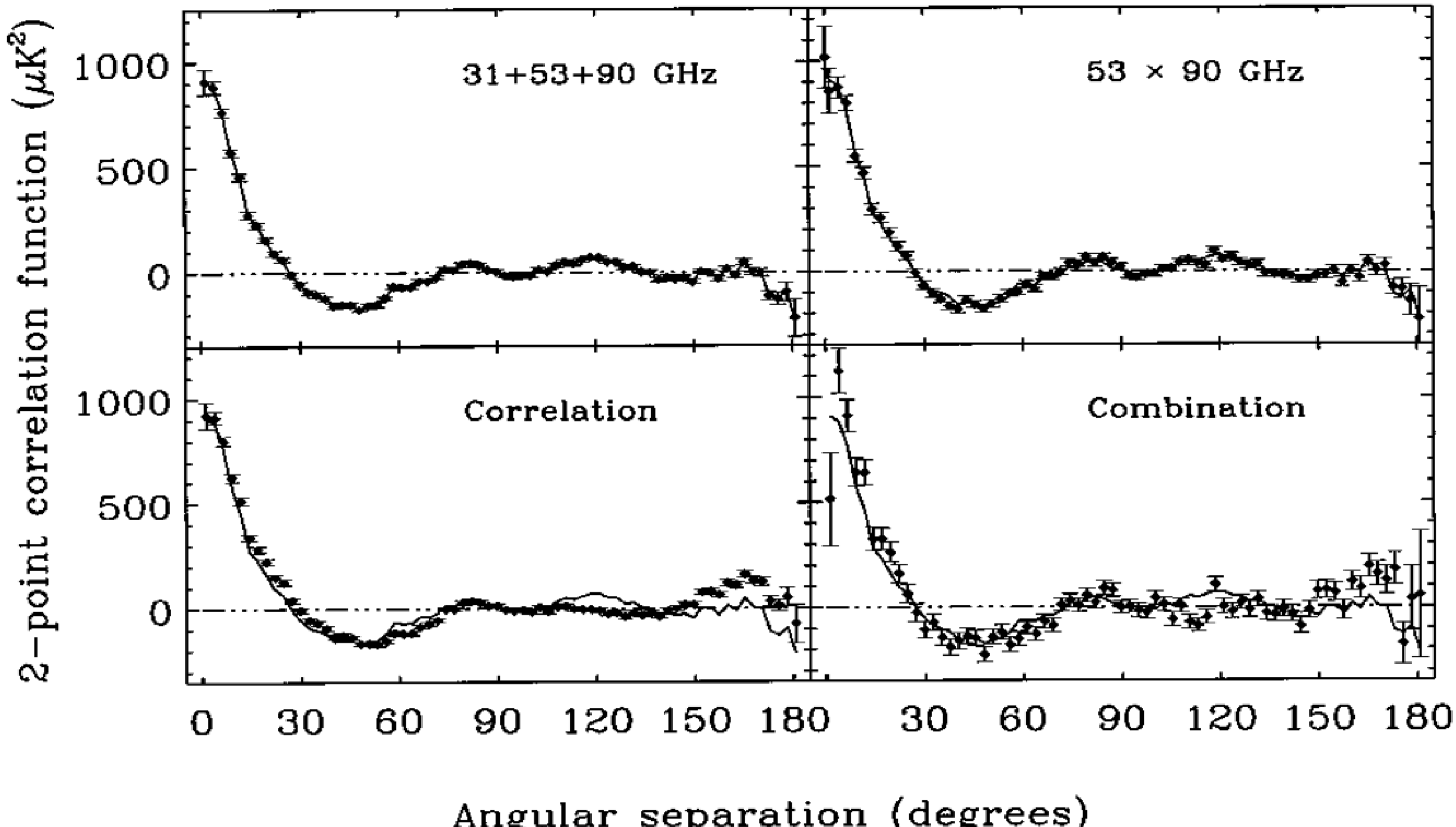
NASA WMAP
Science Team
WMAP 1

THE ASTROPHYSICAL JOURNAL, 464:L25–L28, 1996 June 10

TWO-POINT CORRELATIONS IN THE *COBE*[1] DMR FOUR-YEAR ANISOTROPY MAPS

G. HINSHAW,[2,3] A. J. BANDAY,[2,4] C. L. BENNETT,[5] K. M. GÓRSKI,[2,6] A. KOGUT,[2] C. H. LINEWEAVER,[7] G. F. SMOOT,[8] AND E. L. WRIGHT[9]

Received 1996 January 9; accepted 1996 March 21



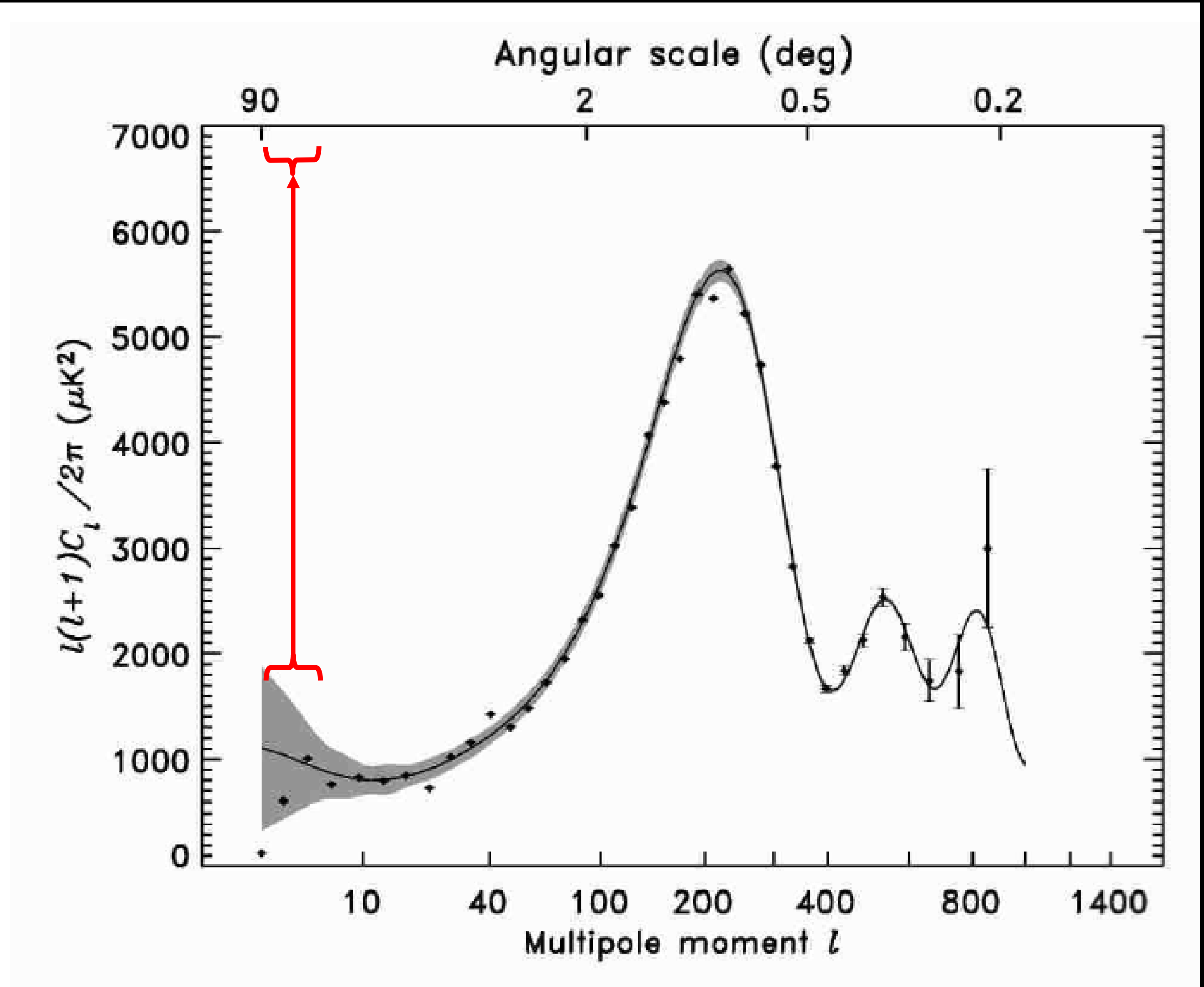
Angular Correlation Function $C(\theta)$

$C(\theta) = \langle T(\Omega_1)T(\Omega_2)\rangle_{\Omega_1 . \Omega_2 = \cos\theta}$

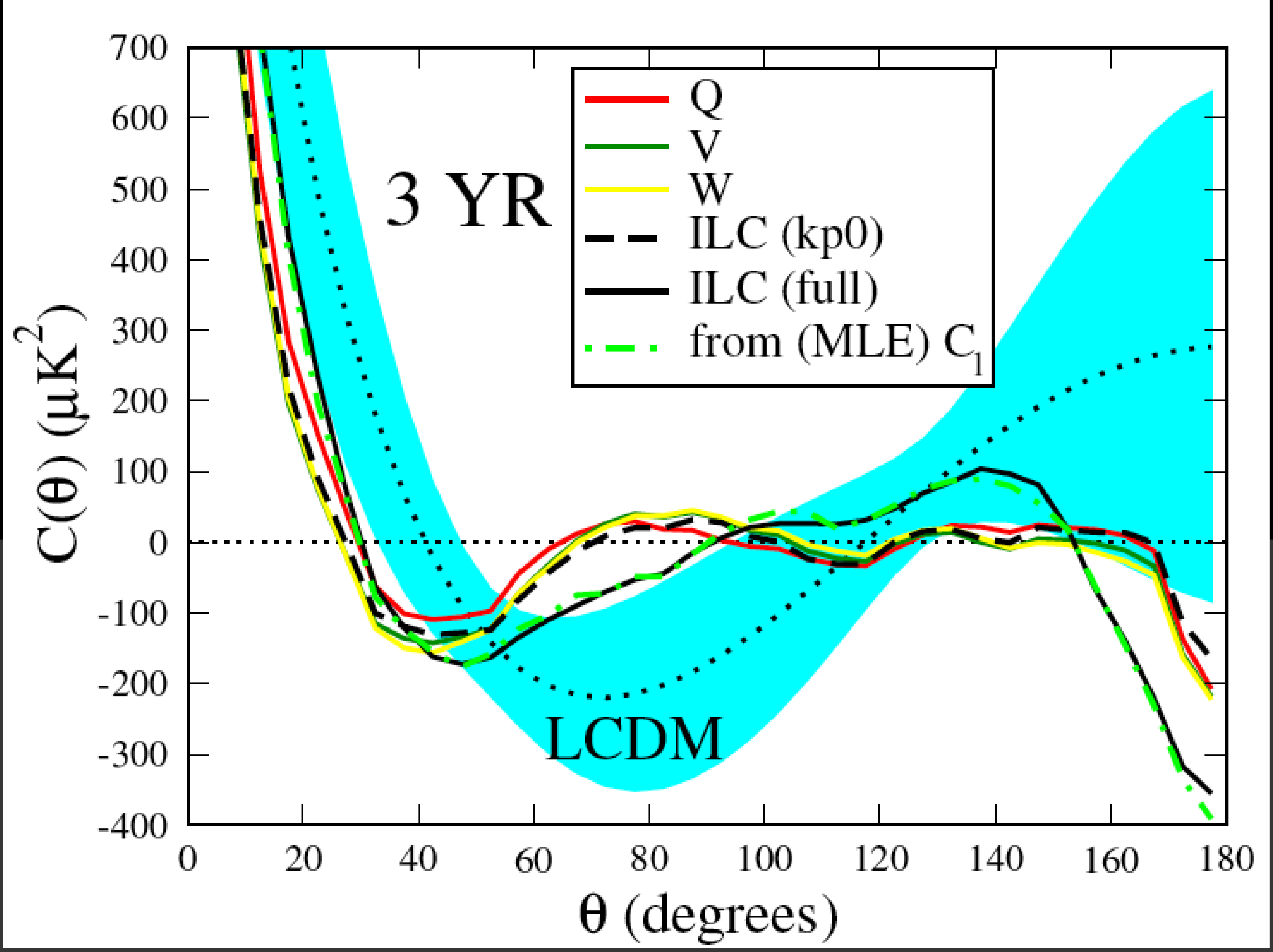
But $C(\theta) = \sum_l C_l P_l(\cos(\theta))$

$\Rightarrow$ Same information as C_l, just differently organized

“The Large-Angle Anomaly”



Two-point angular correlation function



Is the lack of correlations significant?

One measure (WMAP1):

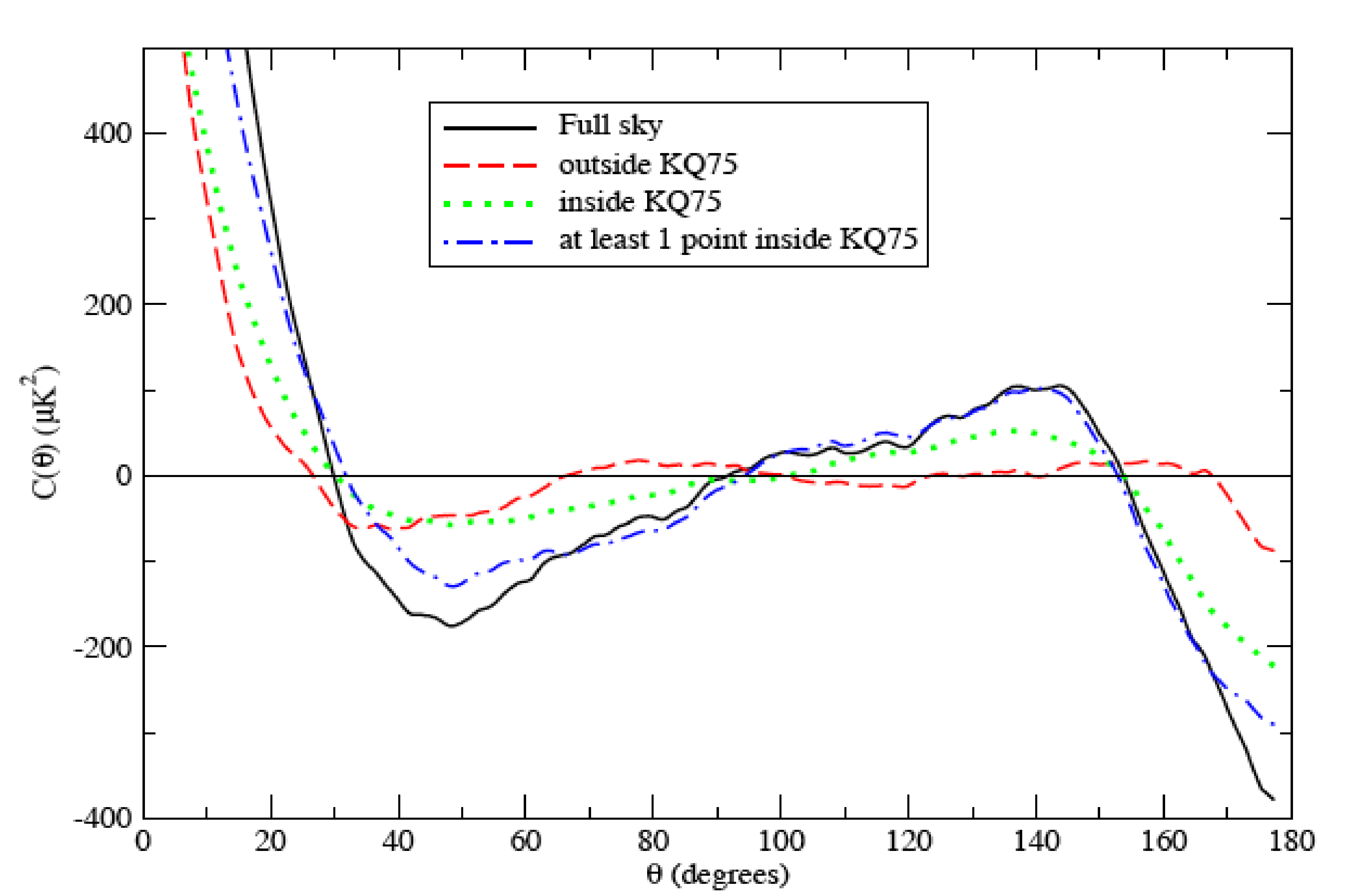
$$S_{1/2} = \int_{-1}^{1/2} [C(\theta)]^2 \, d\cos\theta$$

WMAP statistics of C(θ)

Table 1. The C_ℓ calculated from $\mathcal{C}(\theta)$ for the various data maps. The WMAP (pseudo and reported MLE) and best-fit theory C_ℓ are included for reference in the bottom five rows.

Data Source	$S_{1/2}$ $(\mu\mathrm{K})^4$	$P(S_{1/2})$ (per cent)	$6C_2/2\pi$ $(\mu\mathrm{K})^2$	$12C_3/2\pi$ $(\mu\mathrm{K})^2$	$20C_4/2\pi$ $(\mu\mathrm{K})^2$	$30C_5/2\pi$ $(\mu\mathrm{K})^2$
V3 (kp0, DQ)	1288	0.04	77	410	762	1254
W3 (kp0, DQ)	1322	0.04	68	450	771	1302
ILC3 (kp0, DQ)	1026	0.017	128	442	762	1180
ILC3 (kp0), $\mathcal{C}(>60^\circ)=0$	0	—	84	394	875	1135
ILC3 (full, DQ)	8413	4.9	239	1051	756	1588
V5 (KQ75)	1346	0.042	60	339	745	1248
W5 (KQ75)	1330	0.038	47	379	752	1287
V5 (KQ75, DQ)	1304	0.037	77	340	746	1249
W5 (KQ75, DQ)	1284	0.034	59	379	753	1289
ILC5 (KQ75)	1146	0.025	81	320	769	1156
ILC5 (KQ75, DQ)	1152	0.025	95	320	768	1158
ILC5 (full, DQ)	8583	5.1	253	1052	730	1590
WMAP3 pseudo-C_ℓ	2093	0.18	120	602	701	1346
WMAP3 MLE C_ℓ	8334	4.2	211	1041	731	1521
Theory3 C_ℓ	52857	43	1250	1143	1051	981
WMAP5 C_ℓ	8833	4.6	213	1039	674	1527
Theory5 C_ℓ	49096	41	1207	1114	1031	968

Origin of C(θ)



Full sky
outside KQ75
inside KQ75
at least 1 point inside KQ75
400
200
0
-200
-400
0
20
40
60
80
100
120
140
160
180
C(θ) (μK²)
θ (degrees)

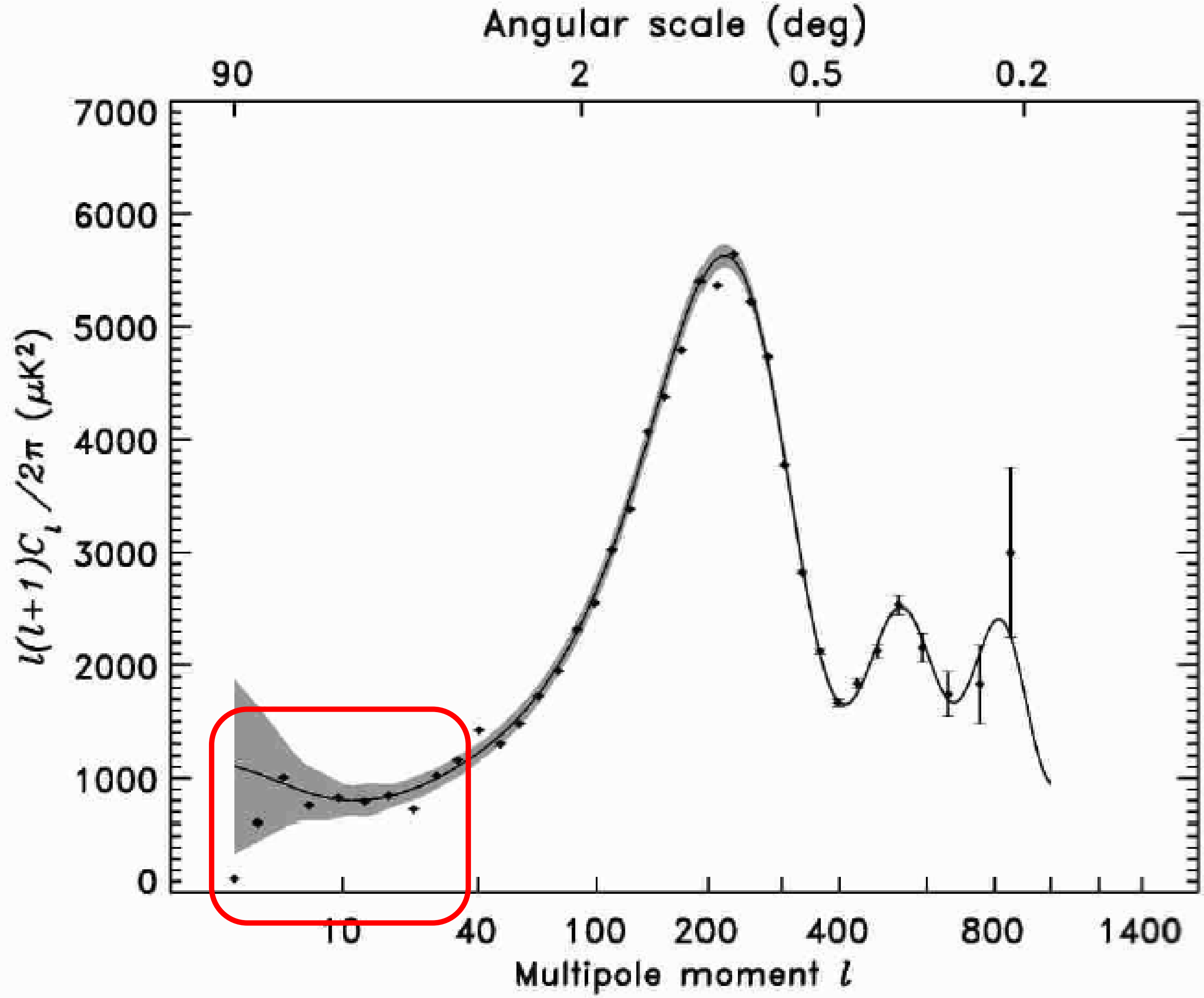
Did this change in Planck?

No!

Statistics of $C(\theta)$

- 0.03-0.1% of realizations of the concordance model of inflationary ΛCDM have so little **cut-sky** large-angle correlation !

and most of those have all low-ℓ C_ℓ small


Angular scale (deg)
90
2
0.5
0.2
7000
6000
5000
4000
3000
2000
1000
0
$l(l+1)C_l/2\pi$ (μK^2)
10
40
100
200
400
800
1400
Multipole moment l

The Conspiracy: minimizing $S_{1/2}$

To obtain $S_{1/2} \lesssim 1000$ with the observed C_ℓ, requires varying C_2, C_3, C_4 & C_5 to minimize $S_{1/2}$!

It’s not the inflaton potential: violation of GRSI

Even if we replaced all the theoretical C_ℓ by their measured values up to ℓ=20, cosmic variance would give only a 3% chance of recovering so little correlation in a particular GRSI realization…
and most of those would be much poorer fits to that theory than is the current data

It's not the inflaton potential:

Even if we replaced all the theoretical C_ℓ by their measured values up to ℓ=20, cosmic variance would give a small chance of recovering so little correlation in a particular realization if C_ℓ are independent … unless

$$\langle (\boldsymbol{C}_\ell - \langle \boldsymbol{C}_\ell \rangle)^2 \rangle \boldsymbol{<<} \langle \boldsymbol{C}_\ell \rangle^{\boldsymbol{2}}$$

$$< S_{1/2} > = \sum_{\ell\ell'} \mathrm{I}_{\ell\ell'} < \boldsymbol{C}_\ell \boldsymbol{C}_{\ell'} > << < S_{1/2} >^{\mathrm{Planck}}$$

$$< \left(S_{1/2} - < S_{1/2} >\right)^2 > \; << \; < S_{1/2} >^2$$

Understanding small $S_{1/2}$

1. ~~"Didn't that go away?"~~
2. "I never believe *a posteori* statistics."
3. "I never believe anything less than 5σ"
4. ~~"Inflation can do that"~~

5. New physics <u>that correlates C_l's</u> (or sets their variances to ~0)

$$\langle C_\ell C_{\ell'} \rangle \not\propto \delta_{\ell\ell'}$$

suggests: $$\left\langle a_{\ell m} a^*_{\ell' m'} \right\rangle \neq C_\ell \delta_{\ell\ell'} \delta_{mm'}$$

Beyond C_ℓ:

Searching for Departures from Gaussianity/Statistical Isotropy

- angular momentum dispersion axes (da Oliveira-Costa, *et al.*)
- BiPoSH coefficients (Souradeep *et al.*)
- cold hot spots, hot cold spots (Larson and Wandelt)
- dipolar modulations
- genus curves (Park)
- hemispherical asymmetries (Eriksen *et al.*, Hansen *et al.*)
- Land & Magueijo scalars/vectors
- multipole vectors (Copi, Huterer, Schwarz, GDS; Weeks; Seljak and Slosar; Dennis)
- parity anomaly
- spherical Mexican-hat wavelets (Vielva *et al.*)
- your favourite technique/anomaly that I missed

Alignments …

Multipole Vectors

Q: What directions are associated w the ℓ^{th} multipole:

$$\Delta T_\ell(\theta,\phi) \equiv \sum_m a_{\ell m} Y_{\ell m}(\theta,\phi) \quad ?$$

Dipole ($\ell = 1$) :

$$\sum_m \mathbf{a}_{1m} \mathbf{Y}_{1m}(\theta,\phi) = A^{(1)}\, \hat{\mathbf{u}}_x^{(1,1)} .(\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

Advantages:

1) $\hat{\mathbf{u}}^{(1,1)}$ is a vector, $A^{(1)}$ is a scalar

2) Only $A^{(1)}$ depends on C_1

Multipole Vectors

General ℓ, write:

$$\sum_m a_{\ell m} Y_{\ell m}(\theta,\phi) \approx$$
$$A^{(\ell)} \left[(\hat{u}^{(\ell,\ell)} \cdot \hat{e}) \ldots (\hat{u}^{(\ell,\ell)} . \hat{e}) \ - \text{all traces} \right]$$

$$\{\{a_{\ell m}, m=-\ell, \ldots, \ell\}, \ell = (0,1,)2, \ldots\} \Rightarrow$$

$$\{A^{(\ell)}, \{\hat{u}^{(\ell,i)}, i = 1, \ldots l\}, \ell = (0,1,)2, \ldots\}$$

Advantages: 1) $\hat{\mathbf{u}}^{(\ell,i)}$ are vectors, $A^{(\ell)}$ is a scalar

2) Only $A^{(\ell)}$ depends on C_ℓ

Maxwell Multipole Vectors

$$\sum_m a_{\ell m} Y_{\ell m}(\theta,\phi)$$
$$= \left[(\mathbf{u}^{(\ell,1)}\cdot\nabla)\ldots(\mathbf{u}^{(\ell,\ell)}\cdot\nabla) r^{-1}\right]_{r=1}$$

J.C. Maxwell,
A Treatise on Electricity and Magnetism, v.1, 1873

Area Vectors

Notice:

- Quadrupole has 2 vectors,

  i.e. quadrupole is a plane
- Octopole has 3 vectors,

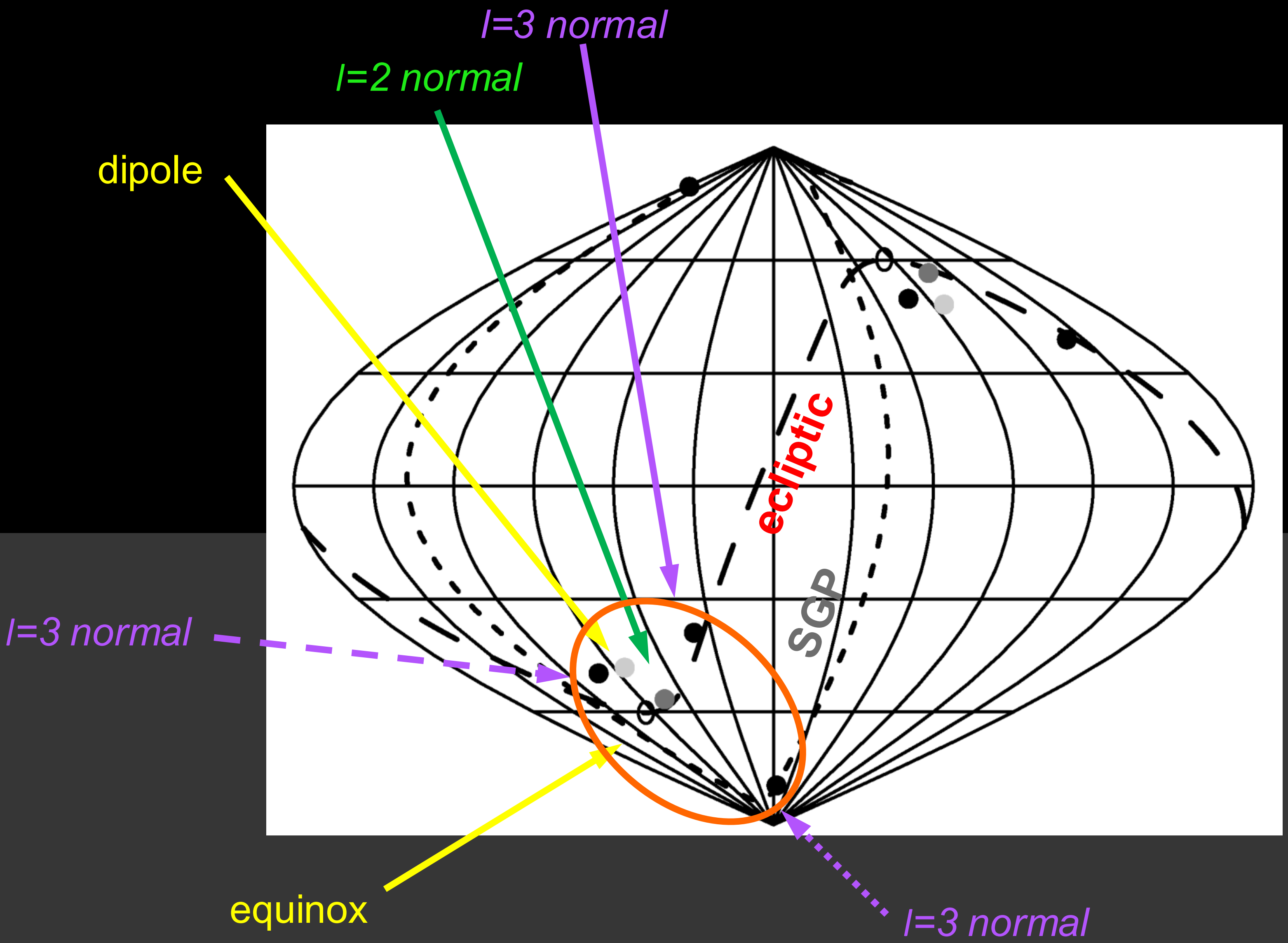
  i.e. octopole is 3 planes

Suggests defining:

$$\mathbf{w}^{(\ell,\mathrm{i},\mathrm{j})} \equiv (\hat{\mathrm{u}}^{(\ell,i)} \times \hat{\mathrm{u}}^{(\ell,j)}) \quad \text{“area vectors”}$$

Carry some, but not all, of the MV information

ℓ =2&3 Area Vectors



Quadrupole plane & 3 octopole planes are aligned with one another

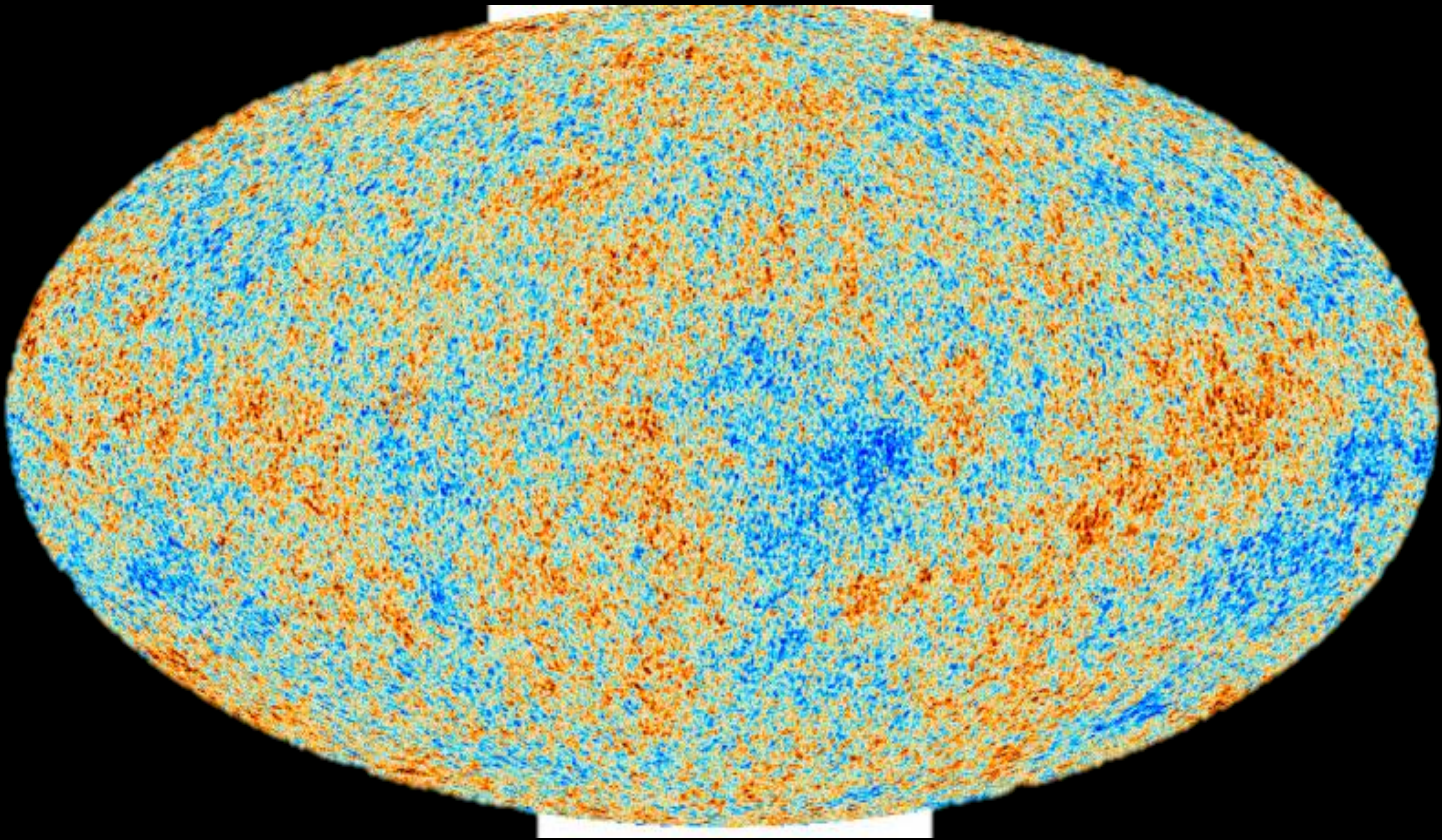
p-value of the quadrupole & octopole planes being so aligned: ~0.1%

Power asymmetry

Dipole modulation

Low Northern Variance

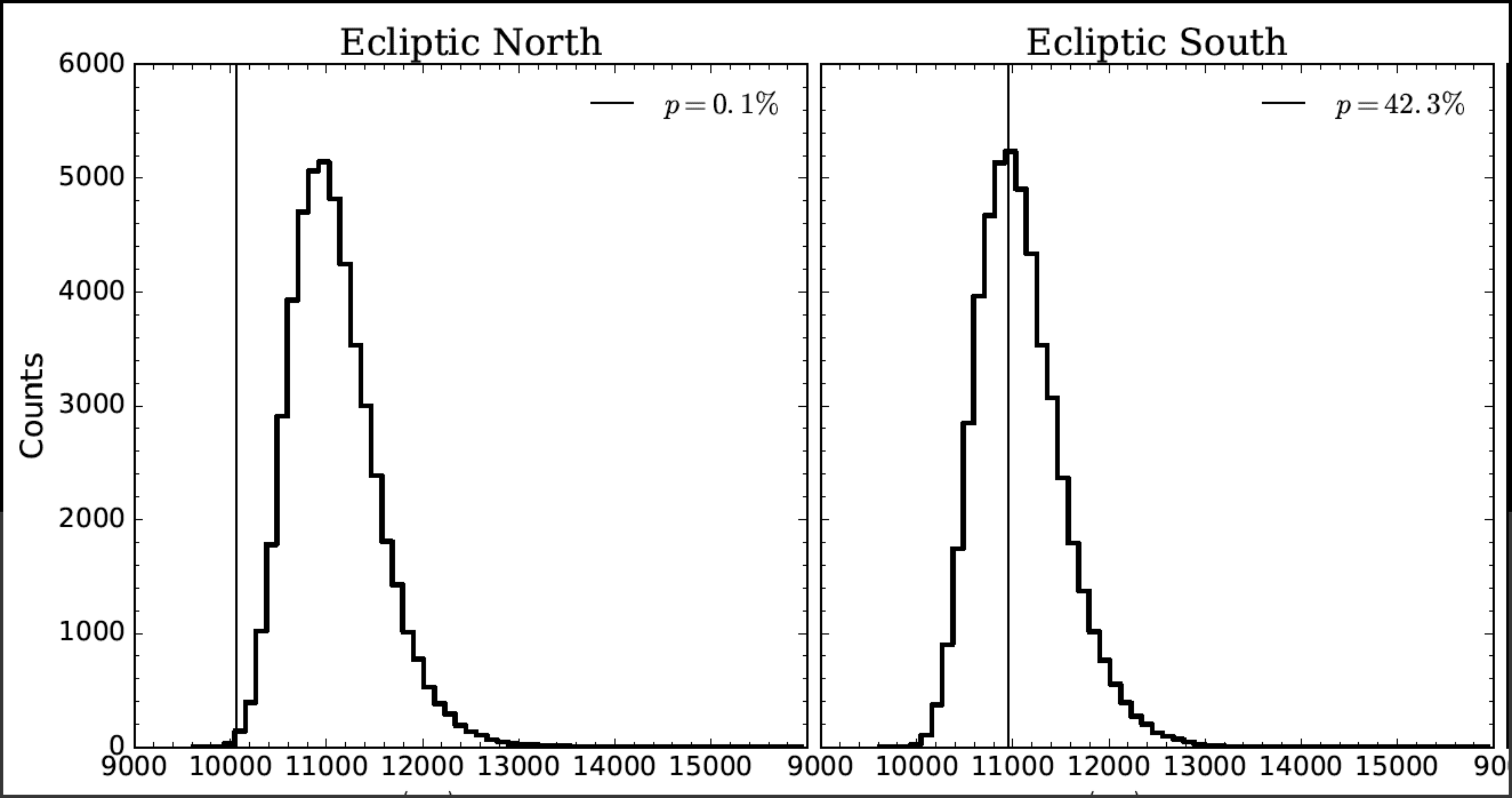
Marcio O'Dwyer (with GDS, Copi, Knox)



Bennett et al 2003

Eriksen et al 2004, and many others

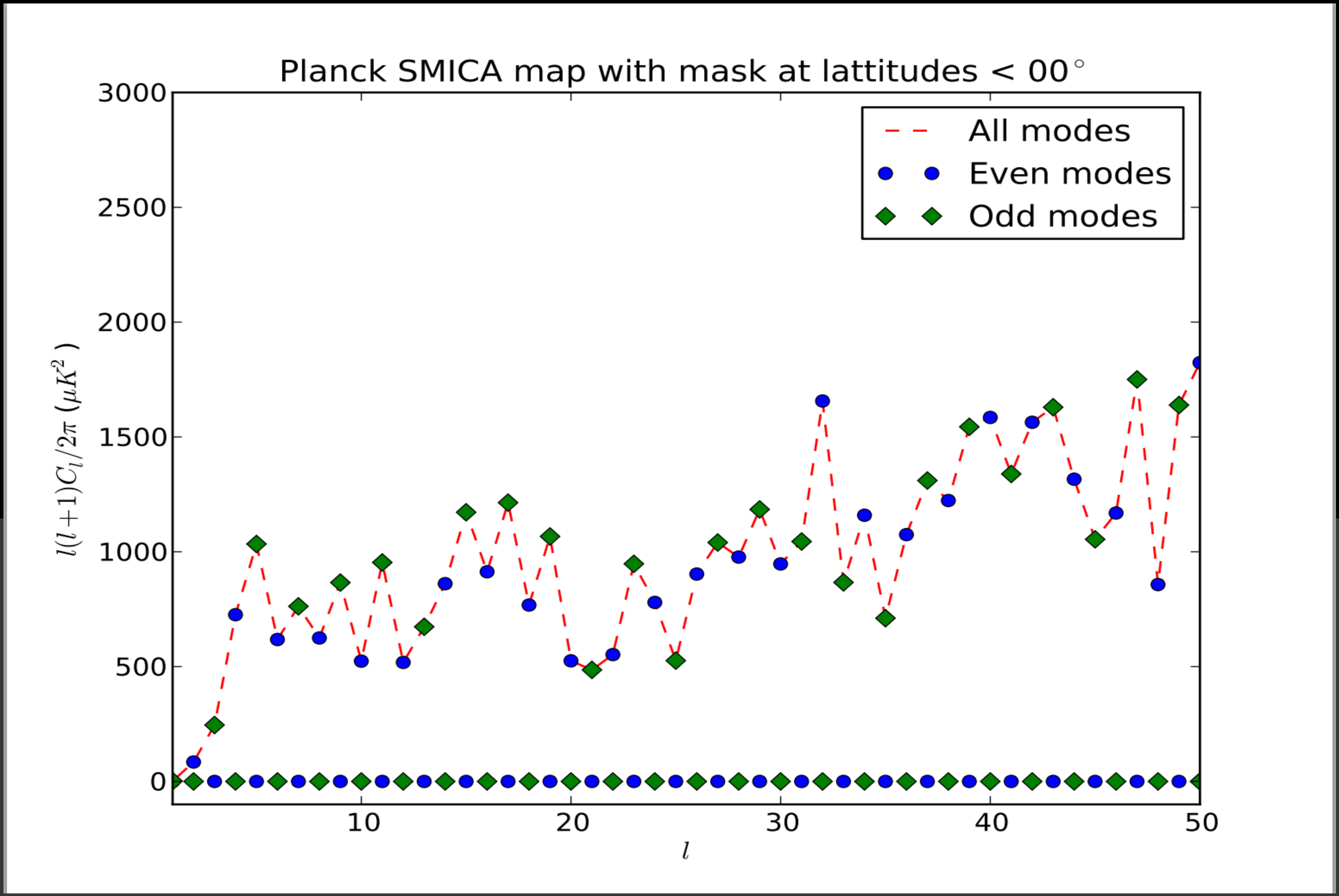
SMICA N vs S variance



p ~ 0.001

Parity anomaly

Parity anomaly



Plot by J. Muir, U. Pittsburgh

And …

With so many anomalies, what do we do?

In preparation: Large-angle anomalies of the CMB and the evidence against statistical isotropy, Physics Reports

[Submitted on 19 Oct 2023]

The Universe is not statistically isotropic

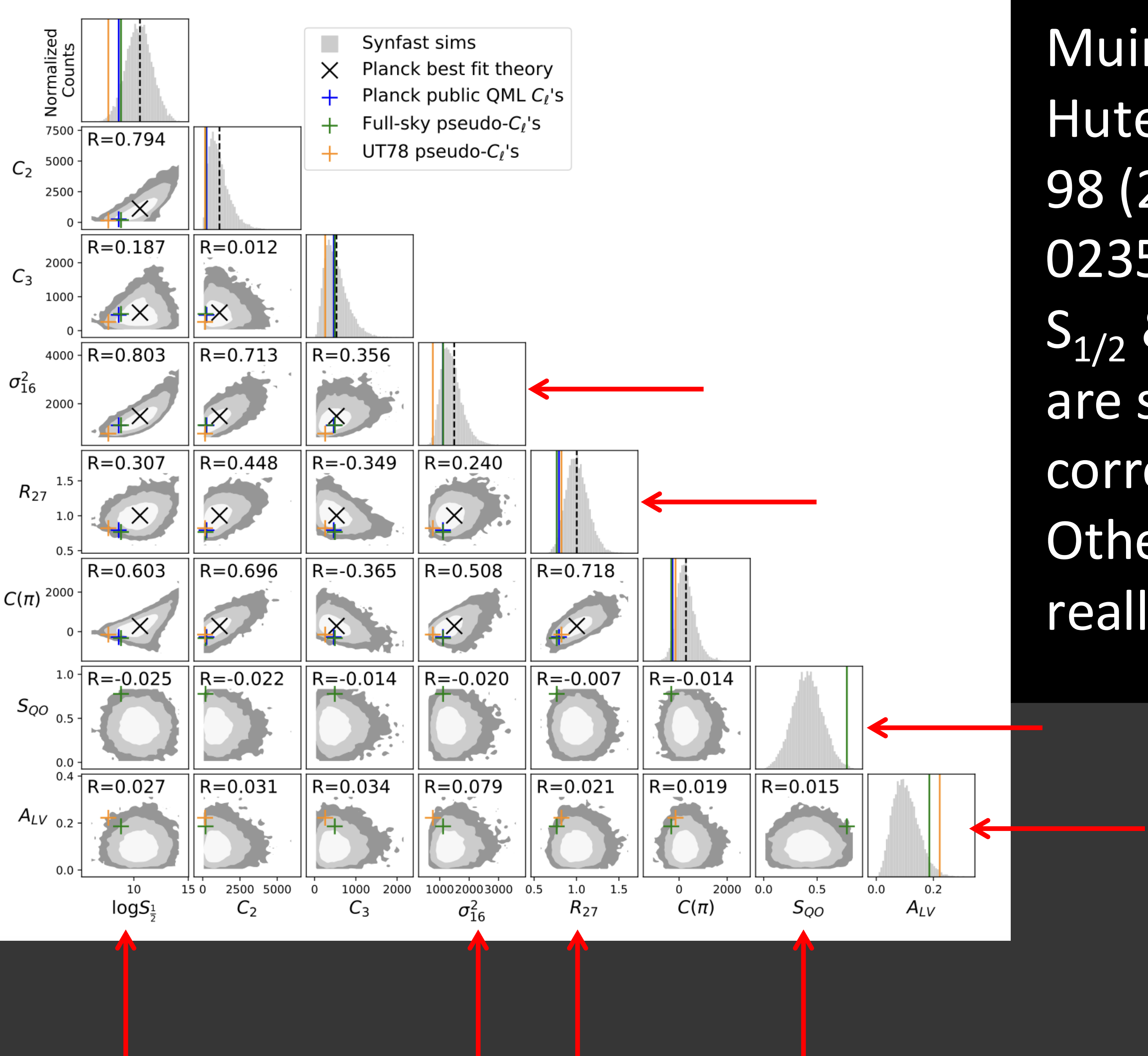
Joann Jones, Craig J. Copi, Glenn D. Starkman, Yashar Akrami

The standard cosmological model predicts statistically isotropic cosmic microwave background (CMB) fluctuations. However, several summary statistics of CMB isotropy have anomalous values, including: the low level of large-angle temperature correlations, $S_{1/2}$; the excess power in odd versus even low-ℓ multipoles, R^{TT}; the (low) variance of large-scale temperature anisotropies in the ecliptic north, but not the south, σ^2_{16}; and the alignment and planarity of the quadrupole and octopole of temperature, S_{QO}. Individually, their low p-values are weak evidence for violation of statistical isotropy. The correlations of the tail values of these statistics have not to this point been studied. We show that the joint probability of all four of these happening by chance in ΛCDM is likely $\leq 3 \times 10^{-8}$. This constitutes more than 5σ evidence for violation of statistical isotropy.

Four “representative” anomaly statistics:

- $\mathbf{S_{1/2}}$ – lack of large-angle correlations, $p \simeq 10^{-3}$
- $\mathbf{R_{TT}}$ – odd-parity preference, $p \simeq 0.01 - 0.05$
- $\boldsymbol{\sigma_{16}^2}$ – low northern variance, $p \simeq (2-4) \times 10^{-3}$
- $\mathbf{S_{QO}}$ – quadrupole-octupole alignment, $p \simeq 4 \times 10^{-(2-4)}$

in Planck 2018 Commander, NILC, SEVEM, SMICA



Muir, Adikhari, Huterer (PRD 98 (2018), 023521) :

$S_{1/2}$ & $\sigma_{16}{}^2$ are somewhat correlated

Others, not really

Four "representative" anomaly statistics:

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in Planck 2018 Commander, NILC, SEVEM, SMICA

But are they correlated in the tails of their pdfs?

But are the anomalies (tails) correlated?

- 10^8 realizations of CMB in best fit LCDM

pairwise correlations

Stat.	Value	$S_{1/2}$	R^{TT}	σ^2_{16}	S_{QO}
			Commander		
$S_{1/2}$	1272	1.5×10^{-3}	$\times 0.6$	$\times 27$	$\times 1.3$
R^{TT}	0.7896	2.8×10^{-5}	3.0×10^{-2}	$\times 1.1$	$\times 1.0$
σ^2_{16}	617.6	1.2×10^{-4}	1.0×10^{-4}	3.1×10^{-3}	$\times 1.7$
S_{QO}	0.7630	8.3×10^{-6}	1.3×10^{-4}	2.3×10^{-5}	4.4×10^{-3}
			NILC		
$S_{1/2}$	1218	1.3×10^{-3}	$\times 0.4$	$\times 29$	$\times 1.3$
R^{TT}	0.7448	4.8×10^{-6}	1.0×10^{-2}	$\times 1.0$	$\times 1.0$
σ^2_{16}	605.9	9.2×10^{-5}	2.4×10^{-5}	2.5×10^{-3}	$\times 1.9$
S_{QO}	0.8203	6.3×10^{-7}	3.8×10^{-6}	1.8×10^{-6}	3.9×10^{-4}
			SEVEM		
$S_{1/2}$	1215	1.3×10^{-3}	$\times 0.8$	$\times 33$	$\times 1.2$
R^{TT}	0.8194	5.6×10^{-5}	5.4×10^{-2}	$\times 1.2$	$\times 1.0$
σ^2_{16}	583.4	6.5×10^{-5}	1.0×10^{-4}	1.6×10^{-3}	$\times 1.5$
S_{QO}	0.6547	6.3×10^{-5}	2.2×10^{-3}	9.8×10^{-5}	4.1×10^{-2}
			SMICA		
$S_{1/2}$	1257	1.4×10^{-3}	$\times 0.6$	$\times 25$	$\times 1.3$
R^{TT}	0.7906	2.8×10^{-5}	3.0×10^{-2}	$\times 1.1$	$\times 1.0$
σ^2_{16}	631.0	1.4×10^{-4}	1.3×10^{-4}	3.9×10^{-3}	$\times 1.8$
S_{QO}	0.8048	1.7×10^{-6}	2.9×10^{-5}	6.6×10^{-6}	9.2×10^{-4}

triplet correlations

	$S_{1/2}$ and σ^2_{16}	S_{QO}
	Commander	
$S_{1/2}$ and σ^2_{16}	1.2×10^{-4}	$\times 1.7$
S_{QO}	9.1×10^{-7}	4.4×10^{-3}
	NILC	
$S_{1/2}$ and σ^2_{16}	9.2×10^{-5}	$\times 0.6$
S_{QO}	2.0×10^{-8}	3.9×10^{-4}
	SEVEM	
$S_{1/2}$ and σ^2_{16}	6.5×10^{-5}	$\times 1.3$
S_{QO}	3.6×10^{-6}	4.1×10^{-2}
	SMICA	
$S_{1/2}$ and σ^2_{16}	1.4×10^{-4}	$\times 2.1$
S_{QO}	2.7×10^{-7}	9.2×10^{-4}

Are the anomalies correlated in LCDM?

Map	p_4	Correlation Factor
Commander	3×10^{-8}	51
NILC	$< 1 \times 10^{-8}$	N/A
SEVEM	18×10^{-8}	40
SMICA	1×10^{-8}	64

Answer: only weakly

Conclusion:

Gaussianity+homogeneity+isotropy is falsified at >5σ in CMB TT correlations!

You can reduce this significance by radical non-Gaussianity (probably + inhomogeneity) but there are additional anomaly statistics that can't be

Discuss:

1. You can't believe data without a model;
i.e., you can't falsify a model without an alternative

2. Look elsewhere penalties
i.e., you can always find anomalous statistics

Louis Lyons "Five Sigma Revisited" CERN Courier 3/7/23:

"There are several reasons why the stringent 5σ rule is used in particle physics.

The first is that it provides some degree of protection against falsely claiming the observation of a discrepancy with the Standard Model. There have been numerous 3σ and 4σ effects in the past that have gone away ...

Why does this happen? Because you notice anomalies and sometimes they're flukes. In other words,

you demand 5σ to mitigate look elsewhere effects! "

Look elsewhere penalties vs.
look more closely rewards

Look elsewhere penalties vs. look more closely rewards

$S_{1/2}$ – lack of large-angle correlations, $p \simeq 10^{-3}$

Look elsewhere:

- **why 60^0? why 180^0? why $C(\theta)^2$? why $d\, cos\theta$?**

Look more closely:

- $p(S_{1/2}^{EE})$ ~ 10^{-3}

Look elsewhere penalties vs. look more closely rewards

$\boldsymbol{\sigma_{16}^2}$ – low northern variance, $p \simeq 3 \times 10^{-3}$

Look elsewhere:

- **why N? why ecliptic? why N_{side}=16?**

Look more closely: (Planck 2013 Isotropy and Statistics)

- **ecliptic not optimum, galactic also ~0.003**
- **p(low (north) skewness, N_{side}=32) = 0.02-0.03**
- **p(high kurtosis, N_{side}=32) = 0.03**

Look elsewhere penalties vs. look more closely rewards

- **R_{TT}** – odd-parity preference, $p \simeq 0.01 - 0.05$

Look elsewhere:

- **why ℓ_{max}=27? why odd>even not even>odd**

Look more closely:

- **first 9 consecutive pairs $C_{2\ell+1} > C_{2\ell}$, $\ell = 1, \ldots, 9$; estimate $p \sim 2^{-9} \approx 0.002$ (x2 for look-elsewhere)**

Look elsewhere penalties vs. look more closely rewards

- **S_{QO}** – quadrupole-octupole alignment, $p \simeq 4 \times 10^{-(2-4)}$

Look elsewhere:

- **??**

Look more closely:

- **“axis of evil” ℓ=2-5 (Land & Magueijo PRL95 (2005) 071301)**
- **“uncanny correlation of azimuthal phases between ℓ=3 & ℓ=5. (*ibid.*)**
- **oriented areas ℓ=2-8 inconsistent at 0.2% (Copi, Huterer, Starkman PRD 70 (2004) 043515)**

Look-elsewhere penalty estimate

- three uncorrelated statistics (SMICA):
  - ($S_{1/2}$&$\boldsymbol{\sigma}_{16}{}^{2}$) (0.00014), S_{QO}(0.0009), P(0.003)
- **most severe penalty -- isotropic** 3d probability to exceed:

$$7.6 \times 10^{-7} \quad \text{i.e. ``}4.8\ \boldsymbol{\sigma}\text{''}$$

“Cherry-picking” penalty?

you need to pay a price for choosing these 4 statistics:

$$\binom{N}{4}$$

where N= the number of statistics that have been considered ?

- You can’t just keep adding statistics that support GRSI because there are an infinite number of them – you’ll never discover anything.
- You should only count uncorrelated statistics
- Huterer et al. chose the 4 statistics they did because there seemed to be a reasonable chance that they were uncorrelated in LCDM, and indeed only $S_{1/2}$&$\boldsymbol{\sigma}_{16}^2$ turned out to be partly correlated,
- They are all “large-angle/low-l” which is a physical selection
- We didn't choose most of the other statistics because:
    a) we expected them to be tightly correlated with the ones we chose.
    b) we couldn't produce much more than 10^8 realizations, and 10^{-8} is already more than $5\boldsymbol{\sigma}$
- There are many likely-to-be-correlated statistics in the “N” (see Planck)
- There are several low-p unlikely-to-be-correlated statistics in the “N” that should be added to the “4”: e.g., low northern skewness, low $S_{1/2}$ of E, dipole modulation alignment out to l>2500

Look-more-closely reward estimate

- $p(S^{EE}_{1/2})$ ~ 10^{-3}
- p(low (north) skewness, N_{side}=32) ~ 0.03
- Extra correlation w ℓ>3 – p~ 0.1
- high l dipole alignment
- cold spot
- first peak dip

- **Collectively << 10^{-5}**

Beyond C_ℓ:

Searching for Departures from Gaussianity/Statistical Isotropy

- angular momentum dispersion axes (da Oliveira-Costa, *et al.*)
- BiPoSH coefficients (Souradeep *et al.*)
- cold hot spots, hot cold spots (Larson and Wandelt)
- dipolar modulations
- genus curves (Park)
- hemispherical asymmetries (Eriksen *et al.*, Hansen *et al.*)
- Land & Magueijo scalars/vectors
- multipole vectors
- parity anomaly
- spherical Mexican-hat wavelets (Vielva *et al.*)
- your favourite technique/anomaly that I missed

Strong evidence that the Universe is NOT (Gaussian random) statistically homogeneous and isotropic

that can be weakened to moderate evidence against SI only if it is radically non-Gaussian (and only until we replace pure C_l statistics by non- C_l statistics) so what?

New Models

New Models

Requirements:

- Break statistical isotropy (including parity)
- Affect scales that were, until recently, causally disconnected

New Models

New models known to explain all/most/several anomalies:

New Models

Physics that breaks isotropy and parity and is <u>already in our theory</u>:

Non-trivial cosmic topology

Stay tuned for Yashar Akrami and Linn Lu

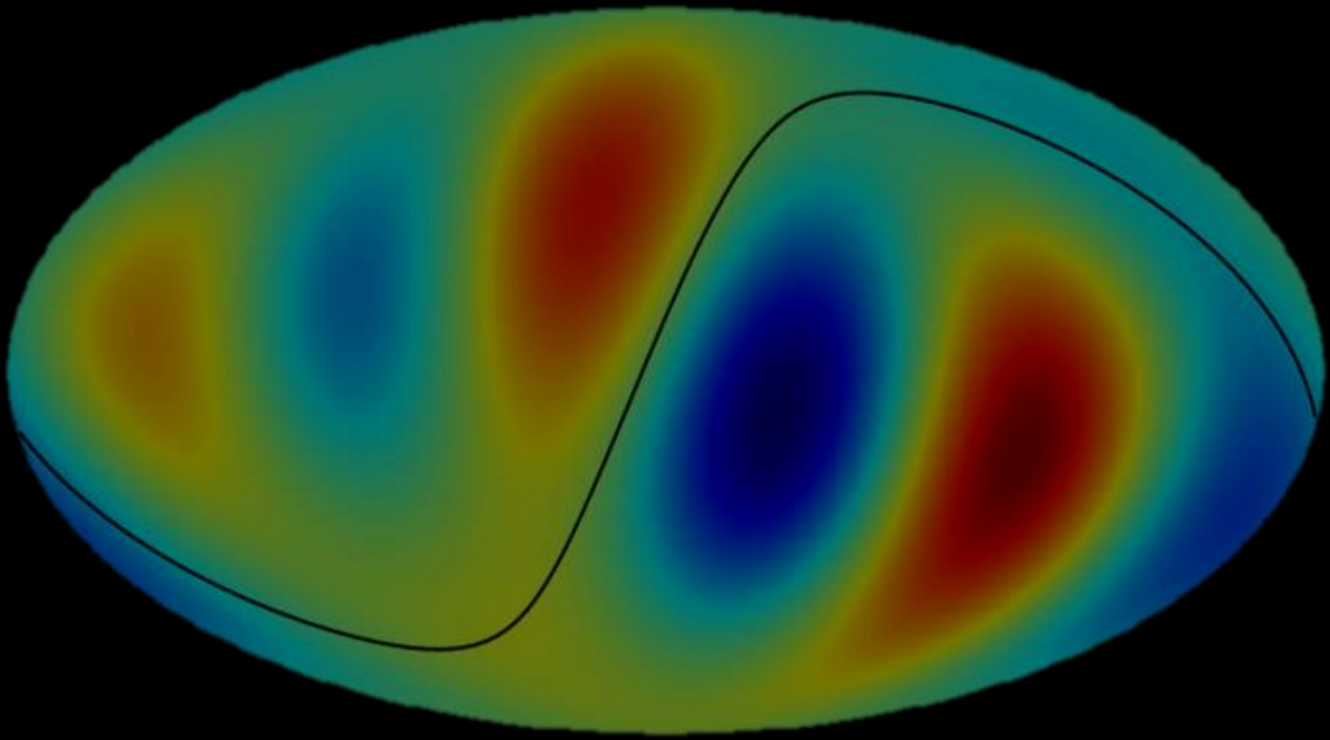
about COMPACT's work WITH Andrew

SUMMARY

The CMB is NOT the realization of a Gaussian random statistically homogeneous & isotropic field.

The Universe is very likely NOT Statistically Isotropic

The cosmic orchestra may be playing a LCDM symphony,
But somebody gave the bass and tuba the wrong score.
They tried hard to keep it quiet. They failed.



We must find a model

to please both Bayesians and frequentists

ABSTRACT DEADLINE

Submit by

JULY 6

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Jun 26 acceptance notice

Jul 6 early registration

Jul 20 late registration

Confirmed speakers include:

Yashar Akrami (IFT UAM-CSIC, Madrid, Spain)
Carlo Baccigalupi (SISSA, Trieste, Italy)
Nicola Bartolo (University of Padova, Padova, Italy)
Angelo Caravano (GRAPPA, Amsterdam, The Netherlands)
Fabio Finelli (INAF OAS, Bologna, Italy)
Anik Halder (IoA and Jesus College, Cambridge, UK)
Niall Jeffrey (UCL, London, UK)
Massimiliano Lattanzi (INFN, Ferrara, Italy)
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Submit your abstract

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